

EDUCATION

University of Engineering and Management

B. Tech (ECE); CGPA (till 5<sup>th</sup> semester): 8.82

Kolkata, India

June 2023 - July 2027

SKILLS SUMMARY

- **Languages:** Python, SQL, JAVA, JavaScript, MATLAB
- **Frameworks:** Pandas, Numpy, Scikit-Learn, Matplotlib, Nodejs, Express.js
- **Tools:** MATLAB, Cadence, Git, Arduino IDE, Supabase
- **Platforms:** PyCharm, Jupyter Notebook, Visual Studio Code
- **Soft Skills:** Rapport Building, Strong Stakeholder management, People Management, Excellent communication

Research Experience

- Spectral ECG Analytics: AI-Driven Early Detection of Sudden Cardiac Arrest | In Press
  - Proposed a machine learning framework for early detection of Sudden Cardiac Arrest (SCA) using ECG signals
  - Extracted frequency-domain spectral features using Welch Power Spectral Density (PSD)
  - Implemented and compared SVM, Decision Tree, KNN, and Neural Network classifiers
  - Achieved classification accuracy up to 95.91% on PhysioNet ECG dataset
- ECG-Based Personalized Workout and Recovery Tracker with Biohaptic Feedback Mechanism | In Press
  - Created a real-time ECG collection system utilizing AD8232 and ESP32.
  - Executed signal filtering and feature extraction based on HRV.
  - Developed closed-loop biohaptic feedback for adjustable exercise management
- VLSI-Enabled Smart Detection of Supraventricular Arrhythmias Using Time Domain Features from Single-Lead ECG | In Press
  - Obtained computationally efficient time-domain attributes (MAV, SSI, RMS) from single-lead ECG recordings.
  - Executed and assessed SVM, KNN, Naïve Bayes, and Logistic Regression classifiers (96.7% maximum accuracy)
  - Created an efficient signal-processing pipeline tailored for low-power embedded/VLSI applications

PROJECTS

- Sick Hen Detection using Computer Vision

June 2025- ongoing

  - Developing a computer vision-based system to identify diseased poultry based on visual and size deviations
  - Implementing object detection to isolate reference objects for distance-aware size normalization
  - Designing logic to estimate relative size variations independent of camera distance
  - Focused on robustness, real-world farm conditions, and deployment feasibility
- Spectral ECG Signal Processing and Classification

ongoing

  - Preprocessed long-duration ECG signals using bandpass filtering to remove noise and baseline drift
  - Segmented ECG data into fixed-length intervals for stable feature extraction
  - Extracted frequency-domain features using FFT and Welch Power Spectral Density methods
  - Built and evaluated machine learning models for classification of normal and SCA conditions
- Mental Health Support Web Application

ongoing

  - Designing a full-stack web application focused on mental health awareness and user well-being
  - Implementing frontend interfaces using HTML, CSS, and JavaScript
  - Using Supabase exclusively for backend services including authentication and database management

CERTIFICATES

- Internet of Things: How Did We Get Here? (University of California San Diego – Coursera)

November 2025

  - Acquired basic knowledge of IoT structure, development, and practical uses in various sectors.
  - Acquired key ideas encompassing embedded systems, sensors, basic networking, and principles of IoT system design
- IoT (Internet of Things) Wireless & Cloud Computing Emerging Technologies (Yonsei University – Coursera)

November 2025

  - Acquired expertise in wireless communication technologies and cloud integration for scalable IoT systems.
  - Investigated new IoT technologies such as cloud systems, networking standards, and contemporary communication frameworks